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### APPARATUS FOR USE IN A SURGICAL PROCEDURE

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#### Abstract

A tapered sleeve is provided that includes a lubricating coating on an inner surface. An implant (e.g., a pre-filled silicon breast implant) is introduced into a large end of the sleeve and extruded into a surgical pocket of minimal access incision size through a small-sized end of the apparatus.

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## Background/Summary

RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 17/574,970, filed Jan. 13, 2022, which is a continuation of U.S. patent application Ser. No. 16/249,174, filed Jan. 16, 2019, which is a continuation of U.S. patent application Ser. No. 15/892,301, filed Feb. 8, 2018, now U.S. Pat. No. 10,213,294, which is a continuation of U.S. patent application Ser. No. 15/195,424, filed Jun. 28, 2016, now U.S. Pat. No. 10,136,988, which is a continuation of U.S. patent application Ser. No. 13/802,332, filed Mar. 13, 2013, now U.S. Pat. No. 9,402,713, which is a continuation of U.S. patent application Ser. No. 13/365,065, filed Feb. 2, 2012, now U.S. Pat. No. 8,555,893, which is a continuation of U.S. patent application Ser. No. 12/228,072, filed Dec. 7, 2007, now U.S. Pat. No. 8,211,173, all of which are incorporated by reference herein in their entireties. Non-provisional application Ser. No. 12/228,072 is a conversion of Provisional Application No. 61/005,777, filed Dec. 7, 2007, done by way of a petition under 37 CFR 1.53 (c) (3) granted on Feb. 19, 2009. This application claims the benefit of the Dec. 7, 2007 filing date.

## FIELD OF INVENTION

[0002] This invention is directed to a delivery apparatus for facilitating the insertion of a silicone implant, such as a breast prosthesis, into a surgically developed pocket.

## BACKGROUND OF THE INVENTION

[0003] This invention relates to the placement of silicone implants within a patient's body. Silicone implants have been in worldwide use for a number of years. While marketing of the implants was halted within the United States for over 15 years, the use of silicone implants has resumed within the United States. One problem with silicone implants is that the implants are provided in a filled condition and must be inserted into a surgical pocket. As a result, traditional surgical approaches require the use of larger incisions in comparison to saline implants which can be inserted through small incisions which are later filled in situ with saline.

[0004] While many patients prefer the more natural qualities of silicone, patients remain apprehensive because of the larger incisions and possibility for visible scars which result from silicone implants.

[0005] An additional concern with the use of silicone implants is that the longevity and integrity of the implants can be compromised by the conventional insertion process. A typical insertion process involves hand manipulation by the surgeon of the implant in order to insert it into the surgical pocket. Studies have shown that implant failures are often associated with an area of minor damage to the outer surface of the implant. The damaged areas are believed to correlate to excessive pressure applied by hand manipulation of the implant and/or damage associated with a “nick” of the implant surface by a “touching” injury such as a fingernail or insertion that damages the implant.

[0006] An additional consideration with respect to silicone implants involves the amount of time required to insert the implants. A traditional hand manipulation of an implant into a surgical pocket can take between 10 to 20 minutes per implant for even a highly skilled surgical practitioner.

Typically, hand manipulation of an implant requires the use of a larger incision and would be done with a saline implant. The amount of time required has a direct bearing on the expense of the procedure, the surgical expense reflecting the surgeon's time, the support staff within the operating room, and the amount of time allocated for the surgical procedure. Accordingly, any improvements to reduce the time required for implantation of the silicone implant will have significant cost savings with respect to the surgical procedure.

[0007] Accordingly, there remains room for improvement and variation within the art.

#### SUMMARY OF THE INVENTION

[0008] It is one aspect of at least one of the present embodiments to provide an apparatus and a process for facilitating the delivery of a silicone implant into a surgically developed pocket.

[0009] It is a further aspect of at least one embodiment of the present invention to provide for an apparatus and process for facilitating the placement of a filled silicone implant into a surgical pocket through a surgical incision that is too small for a manual insertion of an implant.

[0010] It is a further aspect of at least one of the present embodiments to provide for an apparatus and process that allows insertion of a silicone implant through a sleeve defining a small diameter outlet into a patient without direct hand manipulation of the implant.

[0011] It is yet a further and more particular aspect of at least one aspect of at least one of the present embodiments to provide for a process and apparatus that allows for a “touchless” insertion of a silicone implant into a surgical pocket.

[0012] These and other features, aspects, and advantages of the present invention will become better understood with reference to the following description and appended claims.

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## Description

#### BRIEF DESCRIPTION OF THE DRAWINGS

[0013] A fully enabling disclosure of the present invention, including the best mode thereof to one of ordinary skill in the art, is set forth more particularly in the remainder of the specification, including reference to the accompanying drawings.

[0014] FIG. 1 is a perspective view of a prosthetic insertion sleeve along with a silicone prosthesis.

[0015] FIG. 2 is a picture of the insertion sleeve with the silicone implant positioned within the sleeve and showing further hand manipulation of the implant via the sleeve.

[0016] FIG. 3 is a view similar to FIG. 2 showing additional manipulation of the prosthesis through the sleeve and illustrating a portion of the prosthesis exiting a tapered end of the sleeve.

#### DESCRIPTION OF THE PREFERRED EMBODIMENT

[0017] Reference will now be made in detail to the embodiments of the invention, one or more examples of which are set forth below. Each example is provided by way of explanation of the invention, not limitation of the invention. In fact, it will be apparent to those skilled in the art that various modifications and variations can be made in the present invention without departing from the scope or spirit of the invention. For instance, features illustrated or described as part of one embodiment can be used on another embodiment to yield a still further embodiment. Thus, it is intended that the present invention cover such modifications and variations as come within the scope of the appended claims and their equivalents. Other objects, features, and aspects of the present invention are disclosed in the following detailed description. It is to be understood by one of ordinary skill in the art that the present discussion is a description of exemplary embodiments only and is not intended as limiting the broader aspects of the present invention, which broader aspects are embodied in the exemplary constructions.

[0018] In describing the various figures herein, the same reference numbers are used throughout to describe the same material, apparatus, or process pathway. To avoid redundancy, detailed descriptions of much of the apparatus once described in relation to a figure is not repeated in the

descriptions of subsequent figures, although such apparatus or process is labeled with the same reference numbers.

[0019] As seen in reference to FIGS. **1-3**, a sleeve **10** is provided having a general conical shape and which defines a first opening **12** at a larger end of the sleeve and a smaller opening **14** along the tapered terminal tip of sleeve **10**. As best seen in reference to FIG. **1**, sleeve **10** may be provided by a material which is sufficiently flexible and which enables the sleeve **10** to assume a flattened configuration for storage and shipping.

[0020] As seen in reference to FIGS. **2** and **3**, when an implant **20**, such as a silicone implant, is placed through opening **12** into the interior of sleeve **10**, sleeve **10** can be manipulated to conform to the shape of the implant **20** as well as to apply pressure to direct the implant **20** along the length of the sleeve and toward opening **14**.

[0021] Preferably, sleeve **10** is of a flexible material. In one embodiment of the invention, sleeve **10** can be provided by a fabric material such as a plastic-containing fabric which is pliable yet resistant to stretching. It is also envisioned that use of a transparent plastic or other suitable polymer material which has sufficient properties including flexibility and non-elasticity may be used. It is believed that there are advantages to using a transparent or semitransparent material to assist the surgeon in proper orientation of the implant **20** within the interior of sleeve **10**. Suitable transparent materials may include Mylar®, plastics made from Tygon® brand of plastics, vinyls, polyvinyl chloride, and other similar materials. One suitable material which is flexible and transparent includes compositions of ethylene and alpha-olefin copolymers such as the compositions used in IV saline bags. Suitable multilayer films and sealed structures are taught in U.S. Pat. Nos. 6,743,523 and 7,267,885 both assigned to Baxter International Inc., and which are incorporated herein by reference. As disclosed therein, suitable films heat sealed to form suitable containers, are transparent with minimal hazing, and can be sterilized using gas sterilization or heat with intact seals and remain sufficiently flexible and pliable for the necessary manipulation described herein.

[0022] Other attributes of sleeve **10** include the ability to provide sleeve **10** as a sterile component. Accordingly, the material must be capable of withstanding at least one of several conventional sterilization techniques such as a steam autoclave, chemical gas sterilization, or irradiation. Additionally, the interior surface of sleeve **10** may preferably have a low coefficient of friction to facilitate passage of the implant **20** through the sleeve **10**. It has been found that using a surgically appropriate lubricant will facilitate passage of the implant **20** through the interior of sleeve **10**. Such lubricants may be applied directly to the implant **20** or the sleeve can be coated with lubricant or supplied pre-coated with a lubricant that is already present within sleeve **10**.

[0023] Since the size of silicone implants **20** may vary in a range from about 150 cc to about 800 cc, the dimensions of opening **14** may be varied to accommodate various size implants **20**. Preferably, sleeve **10** is provided with an opening **14** sized to fit the smallest implant. The tip opening **14** may be enlarged by cutting portions of the sleeve to provide a larger opening. If desired, indicia may be present on the exterior of sleeve **10** to allow the cutting of the sleeve to the proper dimensions for the size of the implant **20**.

[0024] Alternatively, the tip opening **14** can have a diameter sized to allow the largest standard implant of 800 cc to exit the sleeve. According to this invention, it has been found that an opening **14** having a diameter of about 6 cm will allow delivery of the implant into the surgical pocket. As described below, use of an optional tip would allow for a sleeve to be provided having a single opening size to accommodate the largest implant. Selection of an appropriate tip can then be made based upon the size of the implant.

[0025] While it is believed preferable that sleeve **10** be provided from a single structural substrate, it is also recognized that an equivalent device can be provided of a flexible sleeve **10** having a separate inner liner (not illustrated) which may be present within the interior of sleeve **10**. The liner could either be integral with sleeve **10** or may be a separate layer of material manually inserted within the interior of sleeve **10** at the time of use.

[0026] The larger opening **12** of sleeve **10** allows the implant to be placed within the sleeve with little force or manual manipulation. When the implant is within the interior of sleeve **10**, the larger opening may be twisted closed as seen in FIGS. **2** and **3**. Thereafter, the surgeon is able to apply manual pressure via the sleeve to the implant **20**. The surgeon is thus able to apply pressure to the implant, forcing the implant toward the smaller opening **14**. As seen in reference to FIGS. **2** and **3**, the implant can be forced through the small opening **14**.

[0027] As the surgeon is manipulating the implant through sleeve **10**, the opening **14** is placed within the surgical pocket designed for receiving the implant. Accordingly, tip **14** is inserted through an incision associated with the surgical pocket. As the implant is forced through opening **14**, the surgical pocket can be manipulated slightly to create a vacuum that assists in the placement of the implant into the pocket. Additionally, another useful feature of the apparatus and process is that as approximately half of the implant **20** has been exerted through opening **14**, the remainder of the implant will flow through the sleeve without additional manipulation. Accordingly, once the opening **14** is positioned within the surgical pocket, implant **20** can be manipulated so that the prosthesis **20** is forced into the surgical pocket. The surgeon is able to control the positioning and orientation of the implant **20** by proper rotation and positioning of the sleeve **10** containing the implant **20**.

[0028] It has been found that use of a sleeve **10** can greatly reduce the amount of time required for insertion of an implant **20**. It has been found that the step of inserting a simple implant can occur within a timeframe of about 3 to 20 seconds minutes as compared to a time interval of 5 to 15 minutes for a traditional hand manipulation of an implant. Additionally, because the implant can be inserted through a small opening, the size of the surgical incision can be made smaller than would otherwise be required for a silicone implant.

[0029] The use of the sleeve **10** and implant **20** can be used with incisions. For instance, periareolar, trans axillary, intramammary incisions can be used with the above process and apparatus for insertion of an implant.

[0030] In accordance with this invention, it has been found beneficial to initially lubricate the exterior of implant **20** with an appropriate surgical lubricant such as K-Y® brand sterile lubricant. Following lubrication, the lubricated implant **20** is placed within the sleeve **10** and the implant **20** is forced through opening **14** as a pre-lubrication step. Following this pre-lubrication step, the implant can again be placed within the sleeve and subsequently inserted into the patient's surgical pocket.

[0031] It is also envisioned that, depending upon the coefficient of friction of the interior of sleeve **10** and/or any associated liner, it may be possible to provide other types of lubricants, including dry or powdered lubricant products to the interior of sleeve **10**. Such lubricants are activated by being moistened and would provide an alternative to manually coating the prosthesis with a lubricant.

[0032] If desired, sleeve **10** can further define a structural tip (not illustrated) in association with opening **14**. The tip could be provided of a more elastic material that facilitates insertion of the tip into the surgical pocket. For instance, a separate tip could provide for an extension beyond the existing opening **14** and which would have a narrower initial diameter providing a longer tip which may be more easily inserted within the interior of a surgical pocket. The ability of an optional tip to expand allows the implant **20** to pass through sleeve **10** and opening **14** while positioning the exiting implant **20** further within the surgical pocket. The use of the tip is believed beneficial in that it prevents the passage of the implant from extruding the sleeve from the incision. In other words, the tip provides a deeper positioning for the sleeve **10** more accurately directs the placement of the implant within the surgical pocket. The use of an expandable tip may facilitate the insertion time and lessen the learning curve for surgeons who are using the sleeve **10**.

[0033] An important attribute of sleeve **10**, including any optional tip structure **14**, is that the interior surface of sleeve **10**, including opening **14** and any associated tip member, must provide for a smooth and substantially uninterrupted passageway. It is important that any seams that may be formed or abutments between one type of material to another or from opening **14** to an associated

tip must be of a sufficient smoothness such that the surface of the implant **20** is not degraded. Accordingly, it is envisioned that sonically welded seams or the use of a unitary extrusion process is desired for forming appropriate sleeves **10**. In addition, to the extent opening **14** may be “cut to size”, it is important that the material, once cut, not present any cutting artifacts or roughened edges that could damage the implant **20**. Similarly, the identical concerns must be met by any optional tip used with sleeve **10** such that the material of the tip as well as its method of attaching the tip to the sleeve **10** and sleeve opening **14** must not present any potential implant contact surfaces that could result in damage to the implant.

[0034] Although preferred embodiments of the invention have been described using specific terms, devices, and methods, such description is for illustrative purposes only. The words used are words of description rather than of limitation. It is to be understood that changes and variations may be made by those of ordinary skill in the art without departing from the spirit or the scope of the present invention which is set forth in the following claims. In addition, it should be understood that aspects of the various embodiments may be interchanged, both in whole, or in part. Therefore, the spirit and scope of the appended claims should not be limited to the description of the preferred versions contained therein.

## Claims

1. An apparatus for use in a surgical procedure, the apparatus comprising: a flexible sleeve comprising: a first end region and a second end region, the flexible sleeve being tapered such that a width of the second end region is relatively smaller than a width of the first end region; an enlargeable opening for the second end region; an interior surface that forms an interior cavity, the interior cavity being sized to receive a filled breast implant; and wherein the flexible sleeve is manipulatable such that when the filled breast implant is positioned within the interior cavity a manually applicable directional pressure causes the filled breast implant to extrude from the opening of the second end region.
2. The apparatus of claim 1, wherein the flexible sleeve is cuttable to enlarge the opening.
3. The apparatus of claim 1, wherein the flexible sleeve comprises indicia for cutting the flexible sleeve, the indicia indicating dimensions for accommodating different implant sizes.
4. The apparatus of claim 1, wherein the flexible sleeve is configured to be flattened in a first configuration and conical in a second configuration.
5. The apparatus of claim 1, wherein the flexible sleeve has a terminus in the second end region, wherein an entire peripheral edge of the flexible sleeve forms the opening for the second end region, and wherein the terminus is the entire peripheral edge that forms the opening for the second end region.
6. The apparatus of claim 1, wherein the flexible sleeve is twistable to reduce a volume of the interior cavity.
7. The apparatus of claim 1, further comprising the filled breast implant, wherein the first end region comprises an opening through which the filled breast implant is insertable into the interior cavity.
8. The apparatus of claim 1, further comprising the filled breast implant insertable into the interior cavity through the enlargeable opening of the second end region.
9. The apparatus of claim 1, wherein the enlargeable opening of the second end region has a variable dimension.
10. The apparatus of claim 1, further comprising a liner within the interior cavity of the flexible sleeve.
11. The apparatus of claim 1, wherein the flexible sleeve is transparent.
12. The apparatus of claim 11, wherein the flexible sleeve has a terminus in the second end region, an entire peripheral edge of the flexible sleeve forming the opening for the second end region,

wherein the terminus is the entire peripheral edge that forms the opening for the second end region, and wherein the flexible sleeve is twistable to reduce a volume of the interior cavity.

**13.** The apparatus of claim 12, further comprising a liner within the interior cavity of the flexible sleeve.

**14.** A process of inserting a prosthesis into a surgical pocket comprising: holding a conical sleeve defining an interior cavity, the conical sleeve having a first terminus and a second terminus, the first terminus having a larger diameter than the second terminus, the first terminus and the second terminus each defining a respective opening; placing within the interior cavity of the conical sleeve a silicone implant; positioning the second terminus within the surgical pocket; manipulating the conical sleeve and applying pressure to the implant, directing the implant toward the second terminus; and manipulating the implant through the conical sleeve, thereby forcing the implant through the opening of the second terminus, the implant being deposited within the surgical pocket.

**15.** The process according to claim 14, wherein the conical sleeve comprises a lubricant applied to a surface of the interior cavity.

**16.** The process according to claim 14, wherein the step of placing within an interior cavity of the conical sleeve a silicone implant further comprises applying a lubricant to at least one of either an interior surface of the conical sleeve or an exterior of the silicone implant.

**17.** The process according to claim 14, further comprising moistening a dry lubricant on an interior surface of the conical sleeve.

**18.** The process according to claim 17, further comprising cutting the conical sleeve to enlarge the second terminus and the first terminus remains larger than the second terminus.

**19.** An apparatus for use in a surgical procedure, the apparatus comprising: a flexible sleeve comprising: a first end region and a second end region; an enlargeable opening for the second end region; an interior surface that forms an interior cavity, the interior cavity being sized to receive a filled breast implant; and a dry lubricant on the interior surface, the dry lubricant being activatable by moisture, and wherein the flexible sleeve is manipulatable such that, when the filled breast implant is positioned within the interior cavity, a manually applicable directional pressure causes the filled breast implant to extrude from the opening of the second end region.

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